ENT ATLAS

IMAGING

# Imaging conditions: Ear

*Key facts and case images for the core conditions, then the advanced and rare*

· November 2026

IMAGING

# Core conditions

*The common conditions every resident should recognise, each with its key imaging findings, top differentials, a short checklist and representative scans.*

EXTERNAL EAR & TYMPANIC MEMBRANE · TUMOURS

## Malignant External Otitis or Necrotizing External Otitis

### IMAGING FINDINGS

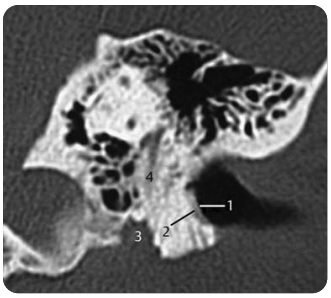
- Soft tissue in the external canal with erosion of its floor and the tympanic bone
- Spread below the skull base: loss of fat at the stylomastoid foramen, parapharyngeal and masticator spaces, temporomandibular joint
- Later: skull base osteomyelitis of the clivus, jugular foramen and petrous apex; venous thrombosis

### TOP DIFFERENTIALS

- Carcinoma of the external canal
- Canal cholesteatoma
- Keratosis obturans

### CHECKLIST

- Facial nerve and lower cranial nerves
- Sigmoid sinus and jugular vein; follow-up MRI changes lag behind clinical cure



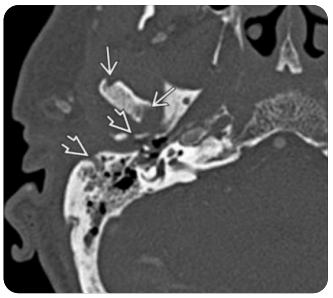

CT, coronal. Necrotizing lesion in the floor of the external auditory canal (1). Clinically, erosive bone lesions (2) with purulent debris and denuded bone…



Axial bone CT shows EAC opacification with focal anterior wall and floor of middle ear erosion in this diabetic patient with painful otorrhea and earl…



Coronal bone CT in the same patient demonstrates anterior EAC wall bony destruction accompanied by complete opacification of the EAC. The middle…



Axial bone CT reveals EAC opacification associated with multiple areas of erosive bony change. The mandibular condyle is also eroded,…

Key facts after Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 1010.

MIDDLE EAR & MASTOID · TRAUMA

# Luxation of the Ossicular Chain

### IMAGING FINDINGS

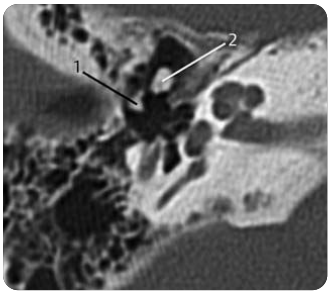
- Separation of the incudostapedial joint; incus rotated or displaced
- Loss of the "ice-cream cone" between malleus head and incus body
- Associated temporal bone fracture through the canal and middle ear

### TOP DIFFERENTIALS

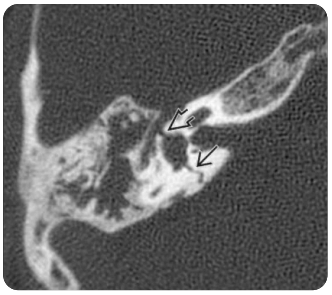
- Ossicular erosion from chronic otitis
- Congenital ossicular malformation
- Partial volume on a single slice

### CHECKLIST

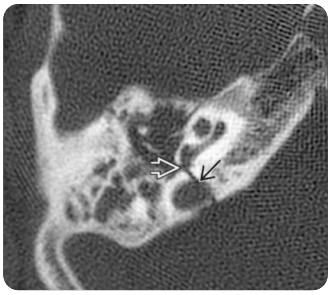
- Compare with the other ear on the same slice
- Check facial canal, tegmen and otic capsule




CT, axial. Traumatic luxation with dislocation of the connection between incus (1) and malleus (2) is seen. The typical icecone configuration of the…



Axial bone CT in a 14-year-old following severe head injury shows a medial transverse, otic capsule-violating fracture traversing the vestibule, with…



Axial bone CT more inferiorly in the same patient shows typical inferior extension of the fracture across the jugular foramen, into the round windo…



Axial bone CT shows a chronic transverse fracture line extending through the otic capsule to the basal turn of the cochlea, where abnormal…

Key facts after Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 968.

MIDDLE EAR & MASTOID · BONE DISEASE

# Otosclerosis, Radiologic Signs around the Oval Niche

### IMAGING FINDINGS

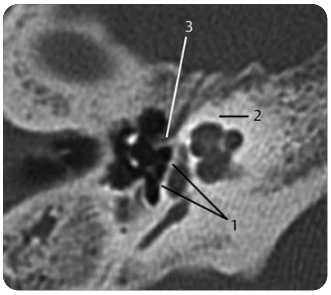
- Lucent focus at the fissula ante fenestram, anterior to the oval window
- Thickened footplate or obliterated oval window niche
- Cochlear form: pericochlear lucent halo ("double ring")

### TOP DIFFERENTIALS

- Tympanosclerosis
- Osteogenesis imperfecta
- Paget disease of the temporal bone

### CHECKLIST

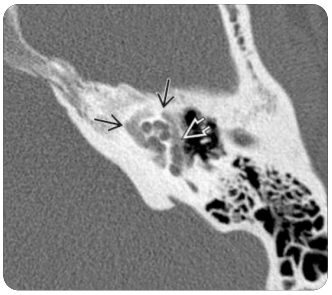
- Compare both ears: usually bilateral
- Report round window involvement and cochlear disease before stapes surgery



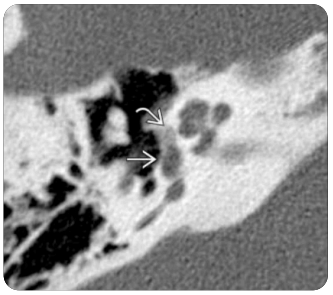

CT, axial. At the edges of the footplate, two densities are visible (1), indicating osseous thickening, which may be considered as otosclerotic foci,…



Coronal right temporal bone CT shows a lytic focus anterior to the oval window, the typical appearance and location of an otospongiotic plaque o…



Axial left temporal bone CT shows cochlear otosclerosis as osteolytic foci surrounding the cochlea. Concurrent fenestral otosclerosis is noted as bon…



Magnified axial right temporal bone CT in a young adult with progressive conductive hearing loss demonstrates typical otospongiotic plaque of…

Key facts after Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 1120.

MIDDLE EAR & MASTOID · TUMOURS

# Glomus Tympanicum (Paraganglioma)

### IMAGING FINDINGS

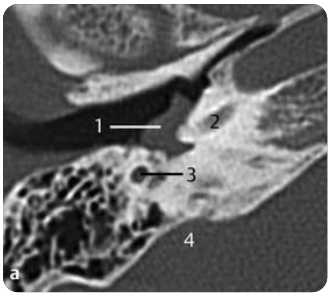
- Small enhancing mass on the cochlear promontory
- Floor of the middle ear intact
- Flow voids only in larger tumours

### TOP DIFFERENTIALS

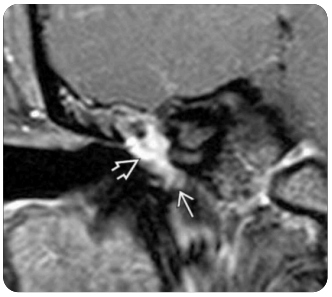
- Glomus jugulare (floor eroded, from the jugular foramen)
- Aberrant internal carotid artery
- Dehiscent jugular bulb

### CHECKLIST

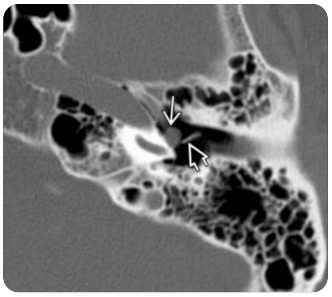
- Check the hypotympanic floor and jugular bulb
- Ossicles and extension into the mastoid



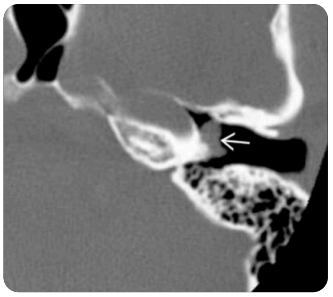

CT, axial. On this axial slice at the level of the hypotympanum, a smooth-bordered lesion (1) is seen, without any destruction of the cochlea (2) or…



Coronal T1 C+ FS MR demonstrates a large GTP filling the middle ear cavity. The floor is intact, separating the tumor from the jugular bulb below.…



An axial bone CT reveals an ovoid glomus tympanicum tumor on the low cochlear promontory abutting the manubrium of the malleus. The…



Axial left temporal bone CT shows a multilobular soft tissue mass on the low cochlear promontory, consistent with a diagnosis of GTP.…

Key facts after Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 1064.

MIDDLE EAR & MASTOID · INFECTION & INFLAMMATION

# Cholesteatoma of the Middle Ear

### IMAGING FINDINGS

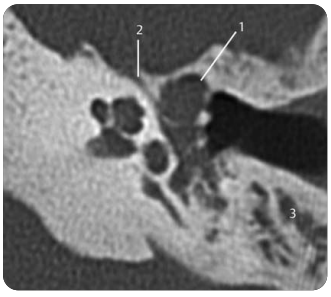
- Soft tissue in Prussak space (attic) or the posterior mesotympanum
- Blunted scutum; eroded, medially displaced ossicles
- Bright on non-EPI diffusion MRI; no enhancement

### TOP DIFFERENTIALS

- Granulation tissue or effusion (no bone erosion)
- Cholesterol granuloma (bright on T1)
- Glomus tympanicum (enhancing, on the promontory)

### CHECKLIST

- Five questions: tegmen, lateral canal fistula, facial canal, sinus tympani, ossicles
- Non-EPI DWI for residual or recurrent disease




CT, axial. The most frequently encountered mass in the middle ear is cholesteatoma, and therefore it is (also) shown in this chapter. In this…



Coronal bone CT reveals a small pars flaccida cholesteatoma filling the Prussak space with blunting of the scutum. The head of the malleus is…



Coronal bone CT shows a large pars flaccida cholesteatoma as a soft tissue mass within the right middle ear and mastoid cavity. There is fistulatio…



Axial bone CT reveals a large cholesteatoma in the middle ear cavity with scalloping of the walls. There is erosion of the anterior portion of the…

Key facts after Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 1050.

For personal study. Follow local protocols and senior advice for patients.

MIDDLE EAR & MASTOID · INFECTION & INFLAMMATION

# Labyrinthine Fistula due to Cholesteatoma

## IMAGING FINDINGS

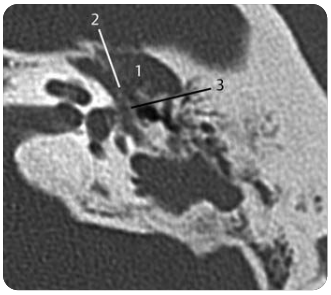
- Loss of the bony cover of the lateral semicircular canal next to cholesteatoma
- Best seen on coronal and Pöschl or Stenvers reformats
- Labyrinthine fluid on T2 may be reduced if labyrinthitis follows

## TOP DIFFERENTIALS

- Partial volume of thin normal bone
- Superior canal dehiscence
- Post-surgical defect

## CHECKLIST

- Report fistula site and size before surgery
- Confirm on thin slices in two planes




CT, axial. Note the smooth-bordered opacification suggestive of cholesteatoma in the anterior part of the middle ear (1), at the level of the…



Coronal bone CT reveals a small pars flaccida cholesteatoma filling the Prussak space with blunting of the scutum. The head of the malleus is…



Coronal bone CT shows a large pars flaccida cholesteatoma as a soft tissue mass within the right middle ear and mastoid cavity. There is fistulatio…



Axial bone CT reveals a large cholesteatoma in the middle ear cavity with scalloping of the walls. There is erosion of the anterior portion of the…

Key facts after Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 1050.

MIDDLE EAR & MASTOID · CONGENITAL

# Congenital Cholesteatoma

## IMAGING FINDINGS

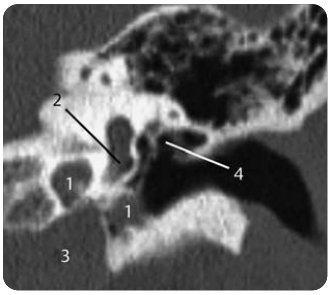
- Round, well-defined pearl behind an intact drum, usually anterosuperior mesotympanum
- Ossicles displaced or eroded, especially the incus long process and stapes
- Bright on non-EPI DWI; no enhancement

## TOP DIFFERENTIALS

- Acquired cholesteatoma
- Glomus tympanicum
- Facial nerve schwannoma of the tympanic segment

## CHECKLIST

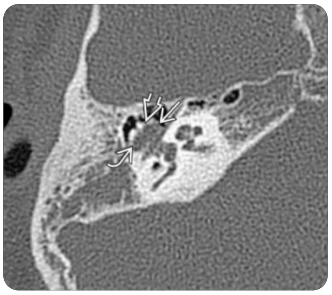
- Intact drum and no history of otitis
- Look for mastoid and petrous apex extension



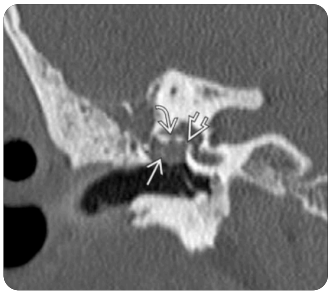

CT, coronal. CT excluded deeper invagination, but revealed other pathology. Coincidentally, an expansive infralabyrinthine…



Coronal temporal bone CT of the right ear demonstrates a large congenital cholesteatoma filling the middle ear cavity with subtle long process of…



Axial bone CT of the right ear reveals a medial epitympanic congenital cholesteatoma eroding the medial head of the malleus and short proces…



Coronal bone CT in the same patient shows CMEC eroding the long process of the incus and filling the oval window niche. The tympanic segment of CNVI…

Key facts after Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 1026.

TEMPORAL BONE & SKULL BASE · TRAUMA

# Skull Base Fractures

### IMAGING FINDINGS

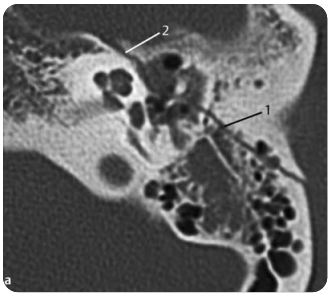
- Fracture line through the temporal bone or skull base; air in the middle ear or cranium
- Otic capsule sparing (most) or violating
- Fluid in the middle ear and mastoid

### TOP DIFFERENTIALS

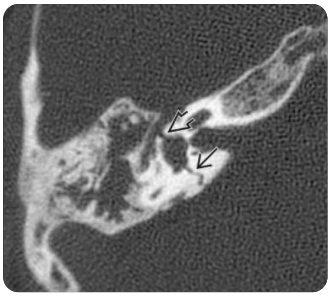
- Sutures, fissures and small canals (pseudofractures)
- Vascular grooves

### CHECKLIST

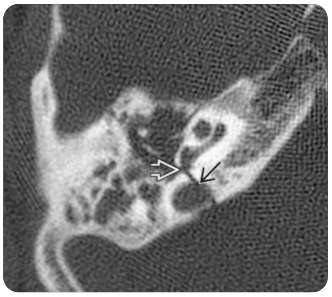
- Ossicles, facial canal (geniculate), tegmen and CSF leak
- Carotid canal involved: CT angiography




CT, axial. Patient with head trauma showing a fracture line through the mastoid (1) extending just anterior to the cochlea (2). Luxation of the…



Axial bone CT in a 14-year-old following severe head injury shows a medial transverse, otic capsule-violating fracture traversing the vestibule, with…



Axial bone CT more inferiorly in the same patient shows typical inferior extension of the fracture across the jugular foramen, into the round windo…



Axial bone CT shows a chronic transverse fracture line extending through the otic capsule to the basal turn of the cochlea, where abnormal…

Key facts after Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 968.

INNER EAR & IAC · CONGENITAL

# Malformations of the Inner Ear

### IMAGING FINDINGS

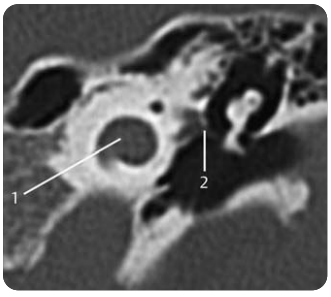
- Abnormal cochlea, vestibule or canals on CT and high-resolution T2
- IP-II: cystic apex, normal basal turn, large vestibular aqueduct
- Common cavity, cochlear hypoplasia, IP-I and IP-III by appearance

### TOP DIFFERENTIALS

- Labyrinthitis ossificans
- Normal variant of canal size

### CHECKLIST

- Cochlear nerve present on oblique sagittal T2
- Cochlear nerve canal width; gusher risk (IP-I, IP-III)




CT, coronal. This coronal view of the cochlea does not show any bony modiolus (1). This empty cochlea is suggestive of an incomplete partition…

Key facts after Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 1102.

INNER EAR & IAC · CONGENITAL

# Vestibular Aqueduct, Enlargements

### IMAGING FINDINGS

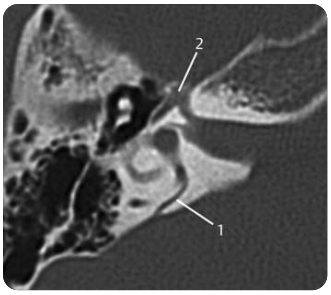
- Vestibular aqueduct wider than about 1 mm at the midpoint or 2 mm at the operculum
- Large endolymphatic sac on T2
- Often with IP-II cochlear deficiency

### TOP DIFFERENTIALS

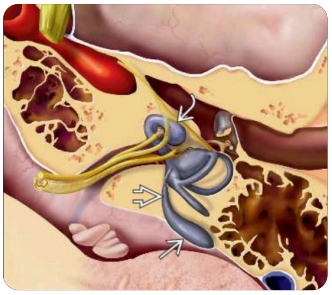
- Normal aqueduct (compare with the posterior canal)
- IP-I cystic cochleovestibular malformation

### CHECKLIST

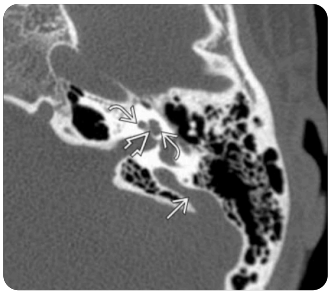
- Look at both ears: often bilateral
- Pendred syndrome association




CT, axial. The normal vestibular aqueduct (1) in its passage from the vestibule to its junction with sinuses in the region of the sigmoid sinus. Ther…



In left inner ear, large endolymphatic sac epidural and intraosseous components are shown. Cochlea is malformed, with absent septation…



Axial temporal bone CT in a 15-year-old boy with sensorineural hearing loss

Koch et al., Diagnostic Imaging: Head and Neck, 3rd ed., p. 1104



Axial temporal bone CT in a 17-year-old girl with Pendred syndrome shows an LVA and absent septation between the middle and apical cochlear turns with…

Key facts after Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 1104.

INNER EAR & IAC · BONE DISEASE

# Labyrinthitis Ossificans (Ossification after Meningitis)

### IMAGING FINDINGS

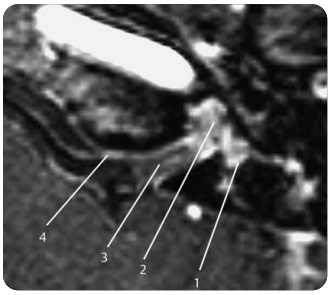
- Loss of the bright fluid signal of the cochlea on high-resolution T2
- Later: bone in the basal turn and canals on CT
- Early: enhancement of the labyrinth

### TOP DIFFERENTIALS

- Labyrinthine or cochlear aplasia
- Cochlear otosclerosis
- Intralabyrinthine schwannoma

### CHECKLIST

- After meningitis: urgent MRI and cochlear implantation
- Report patency of the basal turn




MRI, T1-weighted with gadolinium enhancement, axial. There is enhancement of the vestibule (1), cochlear contents (2), and the…

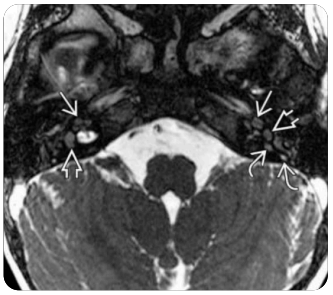


Axial bone CT in a 2-year-old boy with bilateral sensorineural hearing loss

Koch et al., Diagnostic Imaging: Head and Neck, 3rd ed., p. 1116



Axial bone CT in the same patient demonstrates subtotal ossification of the anterior and midportion of the lateral semicircular canal. The…



Axial thin section T2WI MR in a 2-year-old with bacterial meningitis 3 weeks prior shows diffuse, bilateral decrease in normal hyperintense inner ear fluid…

Key facts after Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 1116.

TEMPORAL BONE & SKULL BASE · TUMOURS

# Pseudotumors in the Apex of the Petrous Bone

## IMAGING FINDINGS

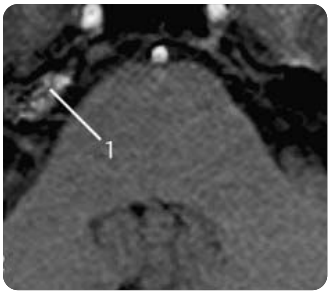
- Asymmetric marrow: bright on T1, suppresses with fat saturation, not expanded
- Trapped fluid: bright on T2, intact septa
- Cholesterol granuloma: bright on T1 and T2, expansile

## TOP DIFFERENTIALS

- Cholesteatoma or epidermoid (DWI bright)
- Petrous apicitis (destruction, enhancement)
- Carotid aneurysm

## CHECKLIST

- Compare with the other apex
- CT for expansion and cortex



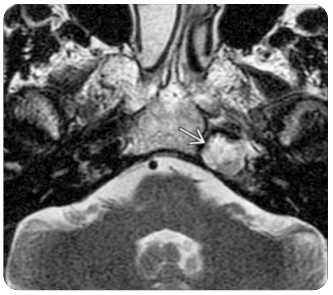

MRI, T1-weighted, axial. T1-weighted image shows that the lesion (1) has high signal intensity compared with brain tissue (2). The contralateral…



Axial bone CT reveals an expansile cholesterol granuloma in the left PA with marginal bone dehiscence present. The right PA is well…



Axial T1WI unenhanced MR in the same patient shows characteristic high signal of PA-CG. If no fat saturation is applied to the enhanced sequences…



Axial T2WI MR in the same patient demonstrates the lesion has inhomogeneous high signal. Cholesterol granuloma contains "old…

Key facts after Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 1152.

INNER EAR & IAC · TUMOURS

# Vestibular Schwannoma

## IMAGING FINDINGS

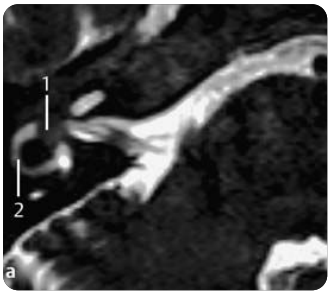
- Enhancing mass centred on the porus, "ice cream on a cone"
- Dark filling defect in bright CSF on high-resolution T2; fundal cap of fluid
- Acute angle with the petrous bone; widened porus; cystic change in larger tumours

## TOP DIFFERENTIALS

- Meningioma (broad dural base, dural tail)
- Epidermoid cyst (DWI bright)
- Facial nerve schwannoma (labyrinthine segment)

## CHECKLIST

- Size in canal and cistern; brainstem and fourth ventricle (Koos)
- Fundal cap and cochlear nerve canal for hearing preservation



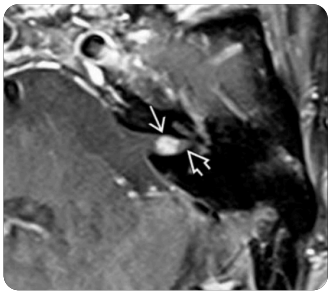

MRI, T2-weighted, axial.A partial lack of fluid in the vestibule and horizontal semicircular canals is evident (1). A mass lesion is present in the vestibul…



Axial T2WI MR reveals a small intracanalicular vestibular schwannoma visualized as a soft tissue intensity mass surrounded by…



Axial T1WI C+ MR demonstrates a large CPAIAC vestibular schwannoma compressing the middle cerebellar peduncle and cerebellar hemisphere.…



Axial T1WI C+ FS MR in a patient with left sensorineural hearing loss shows a small enhancing vestibular schwannoma within the IAC with a 3-…

Key facts after Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 1222.

INNER EAR & IAC · INFECTION & INFLAMMATION

# (Epi)dermoid Cyst

### IMAGING FINDINGS

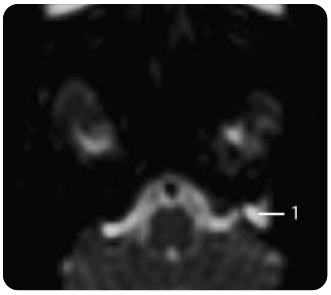
- CSF-like lesion that insinuates around nerves and vessels
- Incomplete suppression on FLAIR
- Bright on diffusion; no enhancement

### TOP DIFFERENTIALS

- Arachnoid cyst (follows CSF on every sequence, DWI dark)
- Cystic schwannoma

### CHECKLIST

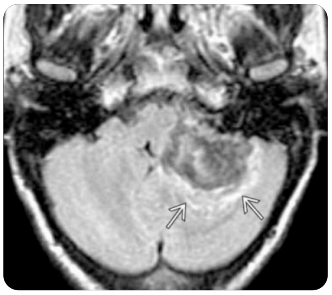
- Encasement of cranial nerves and vessels
- Extent across cisterns before surgery



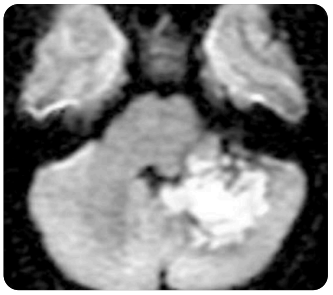

Patient with progressive and asymmetrical perceptive hearing loss.
Dunnebier, Imaging for Otolaryngologists, Fig. 5.9 a



Axial CECT shows a large CPA epidermoid cyst. Note that this non-enhancing low-density lesion appears to invade the left cerebellar…



Axial FLAIR MR of the same patient shows "incomplete" or partial nulling of the signal of this large epidermoid cyst. Associated high signal along th…



Axial DWI MR in the same patient reveals the expected high signal from epidermoid cyst diffusion restriction. DWI sequence allows straightforward…

Key facts after Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 1202.

INNER EAR & IAC · TUMOURS

# Meningioma of the Cerebellopontine Angle

### IMAGING FINDINGS

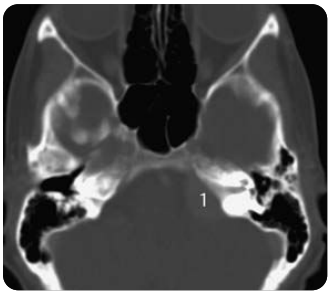
- Broad-based dural mass, obtuse angle with the petrous bone
- Eccentric to the porus; dural tail
- Calcification and hyperostosis; uniform enhancement

### TOP DIFFERENTIALS

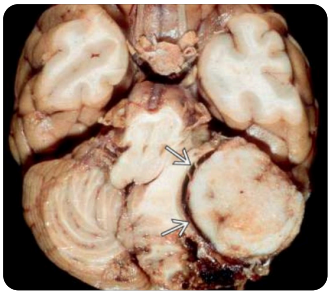
- Vestibular schwannoma
- Epidermoid cyst
- Dural metastasis

### CHECKLIST

- Extension into the IAC, jugular foramen and Meckel cave
- Venous sinus involvement



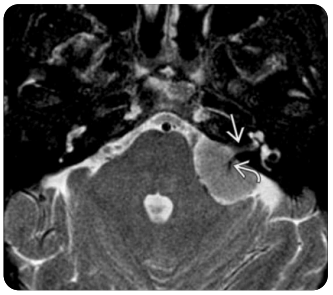

CT, axial. Just before the birth of her child, screening with CT without contrast enhancement suggested a lesion in the left CPA (1). The size of…



Gross pathologic section viewed from below shows a large CPA meningioma with a broad dural base compressing the cerebellum. The specimen…



Axial T1WI C+ FS MR through the IAC shows a meningioma overlying the porus acusticus. Note the dural tail extending along the temporal bone…



Axial T2WI FS MR in the same patient reveals a high-velocity flow void representing a dural artery feeder penetrating the meningioma core. Lo…

Key facts after Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 1228.

TEMPORAL BONE & SKULL BASE · TUMOURS

# Glomus Jugulare Tumor (Paraganglioma)

### IMAGING FINDINGS

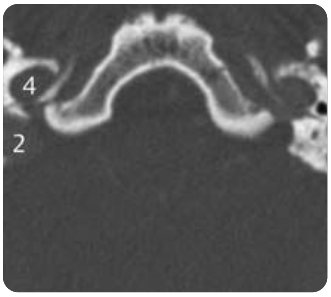
- Jugular foramen mass with permeative, moth-eaten bone erosion
- "Salt and pepper" on T1 with intense enhancement
- Spread superolaterally into the middle ear

### TOP DIFFERENTIALS

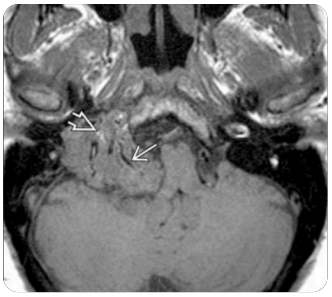
- Jugular foramen schwannoma (smooth scalloping)
- Meningioma (sclerosis, dural tail)
- Dehiscent jugular bulb or metastasis

### CHECKLIST

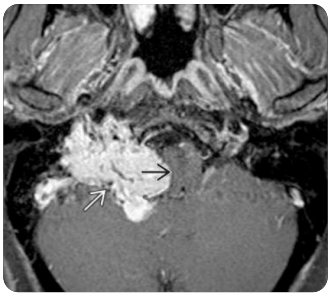
- Carotid, jugular vein and lower cranial nerves
- Multicentric paragangliomas: scan the neck



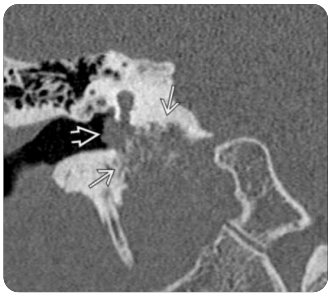

Patient with pulsating tinnitus, without any pathology behind the eardrum.
Dunnebier, Imaging for Otolaryngologists, Fig. 5.20 a



Axial T1WI MR shows a large mass arising from the jugular foramen with multiple areas of salt & pepper. The "salt" represents blood products or…



Axial T1WI C+ FS MR in the same patient shows the classic intense enhancement of this large vascular tumor. There is intracranial extension…



Coronal T-bone CT of the right ear shows the classic permeative-destructive margins of GJP. Note the typical vector of spread superolateral…

Key facts after Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 892.

MIDDLE EAR & MASTOID · INFECTION & INFLAMMATION

# Dural Sinus Thrombosis

### IMAGING FINDINGS

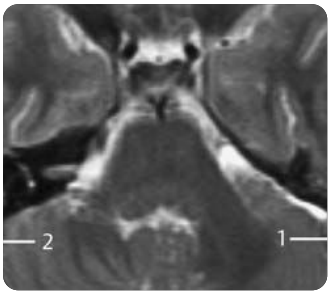
- Filling defect in the sigmoid or transverse sinus on contrast CT or MR venography
- Empty delta sign; dense sinus on non-contrast CT
- Adjacent mastoiditis with bone erosion over the sinus

### TOP DIFFERENTIALS

- Arachnoid granulation (round, CSF signal)
- Hypoplastic transverse sinus
- Slow flow artefact

### CHECKLIST

- Extension to the jugular vein and other sinuses
- Venous infarction and raised intracranial pressure



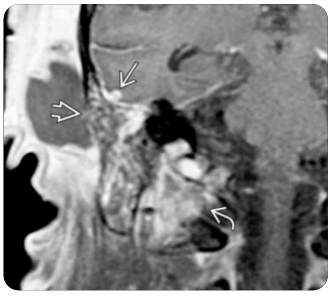

MRI, T2-weighted, axial. Otomastoiditis can spread intracranially, resulting in dural sinus thrombophlebitis, which may compromise the sigmoid and…



Axial CECT in a patient with postauricular tender mass, headache, and fever reveals postauricular abscess and coalescent…



Axial bone CT of the same patient shows loss of mastoid trabecula and a dehiscent sigmoid plate diagnostic of coalescent otomastoiditis. Subtle…



Coronal T1WI C+ FS MR reveals mastoid enhancement with a large periauricular abscess. Lateral mastoid is focally dehiscent with thick and enhancing…

Key facts after Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 1042.

MIDDLE EAR & MASTOID · INFECTION & INFLAMMATION

## Otomastoiditis and its complications

### IMAGING FINDINGS

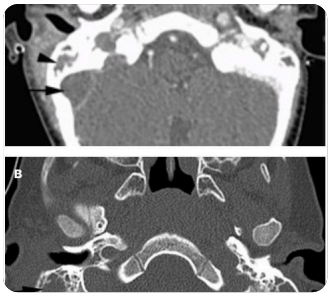
- Opacified middle ear and mastoid
- Coalescence: loss of the mastoid septa and cortical erosion
- Complications: subperiosteal abscess, sigmoid sinus thrombosis, epidural or brain abscess, labyrinthitis

### TOP DIFFERENTIALS

- Uncomplicated effusion (septa intact)
- Cholesteatoma
- Langerhans cell histiocytosis or rhabdomyosarcoma in a child

### CHECKLIST

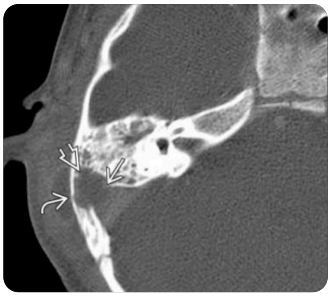
- Contrast CT or MRI with venography when complications are suspected
- Check the sigmoid sinus, tegmen, labyrinth and petrous apex



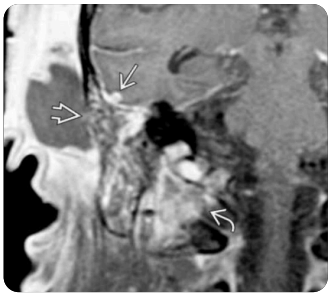
Post-contrast CT shows (a) right coalescent mastoiditis (black arrowhead) and ipsilateral sigmoid sinus thrombosis (black arrow) – bot…



Axial CECT in a patient with postauricular tender mass, headache, and fever reveals postauricular abscess and coalescent…



Axial bone CT of the same patient shows loss of mastoid trabecula and a dehiscent sigmoid plate diagnostic of coalescent otomastoiditis. Subtle…



Coronal T1WI C+ FS MR reveals mastoid enhancement with a large periauricular abscess. Lateral mastoid is focally dehiscent with thick and enhancing…

Key facts after Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 1042.

IMAGING

# Advanced and rare

*Once the core is secure. Each has its full set of images in the app.*

### Inclusion Cholesteatoma and Atresia of the External Auditory Canal

External ear & tympanic membrane · Congenital

Expansion toward and/or destruction of the temporomandibular joint and toward the middle ear structures, abscess formation, and osteomyelitis and (intracranial) spread of infection. · In case of aural atresia (first branchial groove anomaly), other dysmorphic features may be present too, especially in case of syndromal comorbidity.

### Exostoses of the External Auditory Canal

External ear & tympanic membrane · Bone disease

In patients with aural discharge, chronic otitis may be the result of infected epithelial stasis. · In cases with a narrowed orifice and meatus, a meatoplasty might be considered for better aeration and cleaning options. Furthermore, CT might be reassuring, showing a normal aerated middle ear.

### Stenosis of the External Auditory Canal

External ear & tympanic membrane · Bone disease

Beware of accumulation of cerumen or epithelium behind the fibrous wall, and exclude deeper pathology in the middle ear. In most cases, the fibrous wall can be surgically separated from the inner layers of the eardrum without opening the middle ear. · In cases of malignant external otitis or secondary cholesteatoma, deeper pathologies (e.g., to the facial nerve) are of special interest.

### Malignancy of the External Auditory Canal Squamous Cell Carcinoma

External ear & tympanic membrane · Tumours

Extension of bony destruction and features suggestive of infiltrative growth are associated with high morbidity, such as dysfunction of the facial nerve and inner ear. · Adherence to vital structures, such as the internal carotid artery, may preclude surgery.

## Fixation of the Ossicular Chain

Middle ear & mastoid · Bone disease

As mentioned above in differential diagnosis, particular attention must be paid to patients with conductive hearing loss in combination with a normal appearing eardrum and aerated middle ear. · In case of ankylosis of the ossicular chain, accurate evaluation of thin axial and coronal slices may reveal deformities and/or fixation.

## Stapedectomy, Control of Piston Position

Middle ear & mastoid · After surgery

Ossicular prostheses may differ as regards their degree of radiopacity and visibility on CT. For this reason fine slices are required. · Malpositioning of the piston around the incus and possible erosions or resorbtion of the long process and lenticular process due to compression of the piston are difficult to evaluate on CT.

## Osteogenesis Imperfecta

Middle ear & mastoid · Bone disease

In osteogenesis imperfecta affecting the middle ear, fractures of the stapedial crus are often found on middle ear inspection and/or fixation of the footplate in combination with a known (family) history of osteogenesis imperfecta. Note the typical blue sclerae of the eyes in osteogenesis imperfecta type I. · Extension of demineralization, which may be unilateral or bilateral and may vary between both ears. Most cases of osteogenesis imperfecta have minor abnormalities as described in Pathology of the Middle Ear earlier in this chapter. A clinical history of frequent and multiple fractures due to minor trauma everywhere in the body, as well as the typical blue sclerae, might help make the diagnosis.

## Gusher

Inner ear & IAC · Infection & inflammation

A wide communication between the subarachnoid space and the perilymph (a defect at the fundus of the IAC or a dilated cochlear aqueduct) lets CSF pour out when the footplate is opened. · CT clues: a bulbous, widened lateral IAC with a deficient fundus, absent modiolus and a corkscrew cochlea (X-linked incomplete partition type III, POU3F4).

## Persistent Stapedial Artery

Middle ear & mastoid · Variants

An embryonic branch of the ICA (from the hyoid artery) that persists, crossing the promontory and passing through the obturator foramen of the stapes. · CT: a small canal over the promontory, an enlarged anterior tympanic segment of the facial canal, and an absent foramen spinosum (the middle meningeal artery arises from the stapedial artery instead).

## Adenoma of the Middle Ear

Middle ear & mastoid · Tumours

Evaluation of extension to critical structures such as the internal carotid artery and facial nerve is essential when considering whether complete surgical removal is possible.

## Malformations, Treacher Collins

Middle ear & mastoid · Congenital

In deformities of the temporal bone, pay particular attention to the: · External auditory canal, bony or fibrous atresia, appearance and position of the mandibular condyle and temporomandibular joint.

## Fibrous Dysplasia of the Temporal Bone and Skull Base

Temporal bone & skull base · Bone disease

Radiologically on CT, enlargement of bone with intact cortical outlines and a dense ground glass appearance is characteristic of fibrous dysplasia. Fibrous dysplasia incidentally found on MRI may be confused with a malignancy. · In meningioma, bony outlines can be irregular and signs of hyperostosis can be found as well as a dural tail. Sometimes intracranial calcifications might be indicative of an intracranial meningioma.

## Iatrogenic Fenestration

Middle ear & mastoid · After surgery

In fenestrated patients, no actual compression due to masses will be seen. · Beware of suction of the cavity for cleaning. This must be performed very carefully to prevent severe vertigo and inner ear dysfunction.

## Modified Radical Cavity, Persistent Infections

Middle ear & mastoid · After surgery

Attention must be paid to the above-mentioned presence of residual cells. Residual cholesteatoma may present as smooth-bordered, rounded swellings covered by an epithelial or mucosal lining. In cases of destruction of the bony outline to the middle of posterior cranial fossa, a complementary MRI might be helpful for evaluating the extent of the pathology and its relation to intracranial structures. · DWI is especially helpful in discriminating between recurrent or residual cholesteatoma and granulation tissue.

## Osteoradionecrosis

Middle ear & mastoid · Bone disease

Stasis of secretions and infections are frequently the result of eustachian tube dysfunction due to mucosal swelling and loss of ciliary function. Chronic infections may require complete surgical removal of the affected bone and long-term antibiotic treatment.

## Dural Prolapse and Cerebrospinal Fluid Leakage

Middle ear & mastoid · After surgery

Signs of infiltration or expansion. An MRI may be necessary to exclude or evaluate intracranial involvement.

## Hemangioma of the Geniculate Ganglion

Facial nerve · Tumours

Signs of infiltration or expansion. An MRI may be necessary to exclude or evaluate intracranial involvement. · Most tumors, such as schwannoma, demonstrate more sharply defined enhancement.

## Schwannoma of the Facial Nerve

Facial nerve · Tumours

CT and MRI are complementary in the evaluation of the extension of the schwannoma to the middle or posterior cranial fossa as well as to the inner ear structures. Compared with hemangiomas, schwannomas of the facial nerve can be quite large before symptoms occur. · Surgical removal with facial nerve grafting is only considered in (near) total paralysis of the facial nerve or when expansion of the lesion would lead to the damage of surrounding critical structures.

## Endolymphatic Sac Tumour

Inner ear & IAC · Tumours

Signs of bone destruction in the region of the jugular bulb and jugular foramen. – · Cranial nerve VII XI palsies, extension into structures of the neck and skull base.

## Malformations of the Inner Ear and Ossicular Chain

Inner ear & IAC · Congenital

This section illustrates the importance of a systematic comparison of both sides as well as the different compartments of the ear since bilateral congenital pathology may show (subtle) variations between the left and right side.

## Internal Auditory Canal, Enlargements

Inner ear & IAC · Congenital

An internal auditory canal width of less than 3 mm is considered pathologic and may also indicate hypoplasia or absence of its contents. · A width of more than 10 mm may also be pathologic, and assessment of the bony outline of the meatus is essential in the evaluation of pathology. Intralesional calcifications and hyperostosis are suggestive of meningioma.

## Otosclerosis, Retrofenestral

Inner ear & IAC · Bone disease

Extension of demineralization. It may be unilateral or bilateral and may vary between both ears. Most cases of otosclerosis demonstrate minor deformities as described in Pathology of the Middle Ear earlier in this chapter. · In cases of severe bilateral hearing disorders or complete deafness, cochlear implantation may be considered, although the patency of the cochlear lumen is essential for correct and complete electrode insertion. There is a risk of going down a false route due to the severe demineralization and distortion of the normal anatomy.

## Cochlear Implantation

Inner ear & IAC · After surgery

## Schwannoma of the Internal Auditory Canal

Inner ear & IAC · Tumours

Vestibular schwannoma is the most frequent pathology (90%) found in this area. The clinical presentation facilitates the diagnosis because of hearing problems (vestibulocochlear). · Multiple cranial nerve deficits are suggestive of inflammatory processes.

## Cochlear Schwannoma

Inner ear & IAC · Tumours

Subtle and diffuse enhancements are also suggestive of active immunologic pathologies or intralabyrinthine infections. · Fibrosis or ossification may occur over a period of months, excluding the possibility of cochlear implantation in bilateral deafness.

## Schwannoma of the Cerebellopontine Angle

Inner ear & IAC · Tumours

A vestibular schwannoma is the most frequent pathology (90 %) in this area. Its initial clinical presentation will help to determine the origin of the lesion, that is of hearing problems (vestibulocochlear) or facial paresis. · Meningioma may show the so-called dural tail, although this is not always present, and involvement of the internal auditory canal is less common.

## Arachnoid Cysts

Inner ear & IAC · Infection & inflammation

Epidermoid cysts, which are also hypointense on T1-weighted sequences and hyperintense on T2-weighted sequences, can be differentiated with diffusion-weighted imaging (DWI). · In contrast, lipomas are hyperintense on T1-weighted images and hypointense on spin-echo T2-weighted sequences.

For personal study. Follow local protocols and senior advice for patients.

## Cholesteatoma with Intracranial Involvement

Middle ear & mastoid · Infection & inflammation

Epidermoid cysts and cholesterol granulomas are more frequently located in the petrous apex. · Endolymphatic sac tumor originates in the region between the posterior semicircular canal and the middle cranial fossa and sigmoid sinus, with typical imaging patterns (see separate sections).

## Cystic Degeneration in CPA Schwannoma

Inner ear & IAC · Tumours

An arachnoid cyst may be secondary to folding or inclusion of arachnoid membranes by compression from a schwannoma or meningioma. They are mostly located outside the margin of the causative lesion. · A mucocele may be considered when the cyst is in located in a structure covered by mucosa.

## Effects of Radiotherapy on CPA Schwannoma

Inner ear & IAC · After surgery

After stereotactic radiosurgery or fractionated radiotherapy, schwannomas often swell transiently for 6–18 months (pseudoprogression) and lose central enhancement before shrinking. · Do not call failure on early growth alone; follow with serial MRI, and consider treatment failure only with sustained growth over about 2–3 years.

## Neurofibromatosis II (NF II)

Inner ear & IAC · Tumours

Neurofibromatosis may initially present as otologic deficits. – · The only ENT finding in cases of von Hippel–Lindau disease is the presence of a unilateral or bilateral endolymphatic sac tumor.

## Astrocytoma in the Cerebellopontine Angle

Inner ear & IAC · Tumours

Sharpness of the margins helps to differentiate between a benign and malignant lesion. · An endolymphatic sac tumor originating near the osseous mastoid part behind the labyrinth is mostly confined extradurally.

## Labyrinthitis and Subclinical Meningitis

Inner ear & IAC · Infection & inflammation

Otoscopy may exclude middle ear infections although CT may be indicated in some cases. MRI will show (retro)cochlear enhancement in cases of (acute) infections. · Inflammatory involvement of the intralabyrinthine structures may result in fibrosis and ossification. In bilateral cases, this may hamper insertion of a cochlear implant.

## Otosyphilis and Intracranial Complications

Inner ear & IAC · Infection & inflammation

Cranial nerve deficits, rapid progression, and multiple lesions are strongly indicative of inflammatory disorders. If malignancy is suspected, beware of involvement of other organs as primary sites of possible metastases. · Inflammatory involvement of the intralabyrinthine structures may result in fibrosis and/or ossification. In bilateral cases, this may hamper insertion of a cochlear implant, although its success may also be restricted by retrocochlear deficits.

## Infection of the Skull Base and Gradenigo Syndrome

Temporal bone & skull base · Infection & inflammation

In Gradenigo syndrome, the dural branches of the first division of the trigeminal nerve are involved in the symptoms of pain; the abducens nerve is paralyzed because of compression in the Dorello canal in the petrous apex. Pseudo-Gradenigo may be due to a nasopharyngeal carcinoma infiltrating the skull base. · Malignant otitis externa may lead to skull-base infection. Immunocompromised and diabetic patients are prone to bacterial (e.g., Pseudomonas) or fungal (e.g.,) infections. Sequential radiographic and radionuclide (e. Aspergillus g., technetium and gallium) imaging may be used to monitor the efficacy of treatment. Complications may result from the spread of infection to adjacent intracranial or head and neck structures.

## Petrositis

Temporal bone & skull base · Infection & inflammation

Clinical symptoms may help differentiate between inflammatory disorders, asymptomatic pseudotumors, and other pathology. CT and MRI are complementary. Knowing whether the origin is intradural or extradural and the presence of bony changes help to establish the most likely diagnosis.

## Intracranial Abscess from an Otologic Focus

Middle ear & mastoid · Infection & inflammation

Otogenic abscesses lie in the temporal lobe (through the tegmen) or the cerebellum (through Trautmann's triangle or the sigmoid sinus). · MRI: a ring-enhancing lesion with restricted diffusion in the centre (unlike necrotic tumour) and surrounding oedema.

## Intracranial Hemorrhage Due to a Via Falsa

Temporal bone & skull base · After surgery

During mastoidectomy, drilling through the tegmen or the sinodural angle without recognising the middle fossa dura can enter the temporal lobe ('via falsa'). · Find the tegmen early, thin it under irrigation and follow it, keeping the dura in view; a low-lying tegmen is visible on preoperative coronal CT.

## Intracranial Hemorrhage due to a Middle Cranial Fossa Approach

Temporal bone & skull base · After surgery

The middle fossa approach needs temporal lobe retraction, so postoperative imaging may show contusion, extradural or intracerebral haematoma. · CSF leakage through a tegmen defect into a radical cavity shows as fluid in the cavity with a bony gap on coronal CT; MRI or cisternography confirms the dural defect.